

VaaniPay

Voice-First IVR Financial Inclusion Platform

ACM Challenge Week — Hackathon 1.0 | Problem Statement 3: FinTech for Financial Inclusion

Team Information

Team Name: Team VaaniPay

GitHub Repository: <https://github.com/kisnaXD/acm-manipal-vaanipay>

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Problem Statement

PS 3: FinTech for Financial Inclusion: While digital payment systems have grown rapidly, a large section of the population — including gig workers, small business owners, and rural communities — still lacks access to credit, financial planning tools, and protection against fraud. The challenge is to develop a solution that leverages technology to make financial services more accessible, secure, and inclusive, addressing the needs of underserved populations.

Ideation

More than 300M Indians have no access to digital financial services, not for want of need, but for lack of any solution that doesn't presuppose use of a smartphone, internet, and literacy in English. For the hundreds of millions of Indians who use only a basic feature phone or landline to communicate-gig workers, daily laborers, small farmers, rural women-whose financial life is conducted entirely in cash, with zero security in terms of savings, constant exposure to moneylenders, and permanent exclusion from credit, there is not one mainstream financial application. UPI is already the future of payments in urban India, but the 250M Indians who will never be able to afford, or will lack reason to use, a smartphone, have no access to a single app.

VaaniPay is founded on one simple insight: do not ask the underserved Indians to update the infrastructure they use, bring financial services to the infrastructure they already have. A telephone. Any telephone. Using IVR technology, integrated with purpose-built Indian language speech models from Sarvam AI, and built on the NPCI's already extant USSD-UPI rails, VaaniPay allows anyone with a telephone to dial one number, speak naturally in their own language, and access payments, credit, insurance and savings entirely through voice-no smartphone, no internet, no literacy required.

The NPCI's *99# USSD service has already demonstrated the regulatory and technical feasibility of using a universal code (a phone number) for banking, however its current,

keypad driven interface only supports nine languages, and it includes no support for savings, insurance or credit.

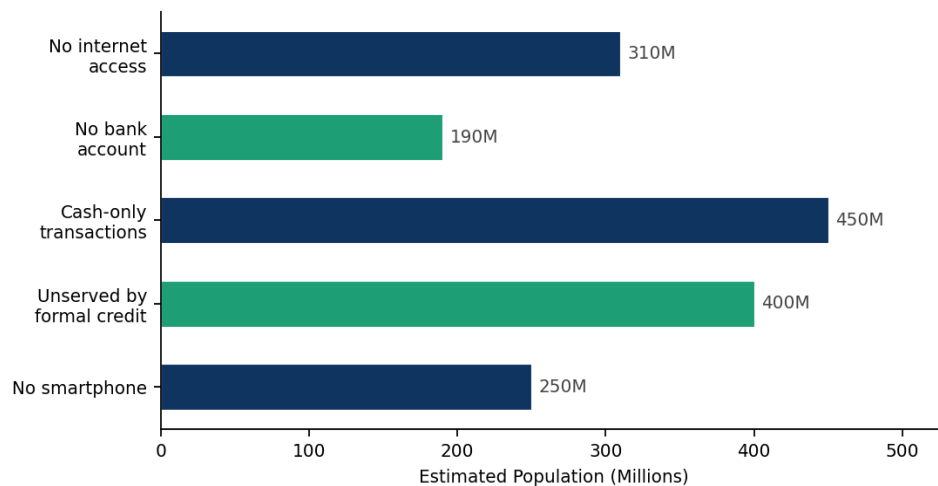


Figure 1: Scale of financial exclusion in India: the problem VaaniPay addresses

Implementation

When the user dials the VaaniPay number, the call is received by the IVR engine that is powered by Twilio or Asterisk. There is no data plan needed on the caller's phone-the call proceeds over the normal PSTN or GSM line. The system automatically detects the user's language from the first spoken phrase by the user, or gives a short language choice if none are detected. The entire audio is streamed to Sarvam AI's STT pipeline, which performs transcription with a word error rate under 12% on the lossy, 8KHz telephony audio.

The transcribed text is then fed to a custom NLP intent parser which is fine-tuned for vocabulary around financial domains across Indic languages. This determines one of the many intents (payment, balance inquiry, loan application, insurance purchase, credit score check, savings deposit etc.) as well as the entities for that intent (amount, recipient, account etc.) for the request. Authentication: Before the banking orchestrator attempts to process the transaction request downstream, it uses the registered voice print along with a spoken 4-digit mPIN in order to verify the identity of the caller.

The banking orchestrator routes the validated request to the corresponding downstream systems (NPCI USSD bridge for UPI payments, NBFC partner APIs for loans, InsurTech partner integrations for insurance). After the transaction is confirmed, the response is then converted into natural speech using Sarvam AI's TTS engine for the specific user's language, and played back over the call. The entire experience from dial to confirmation for a UPI payment takes under 50 seconds.

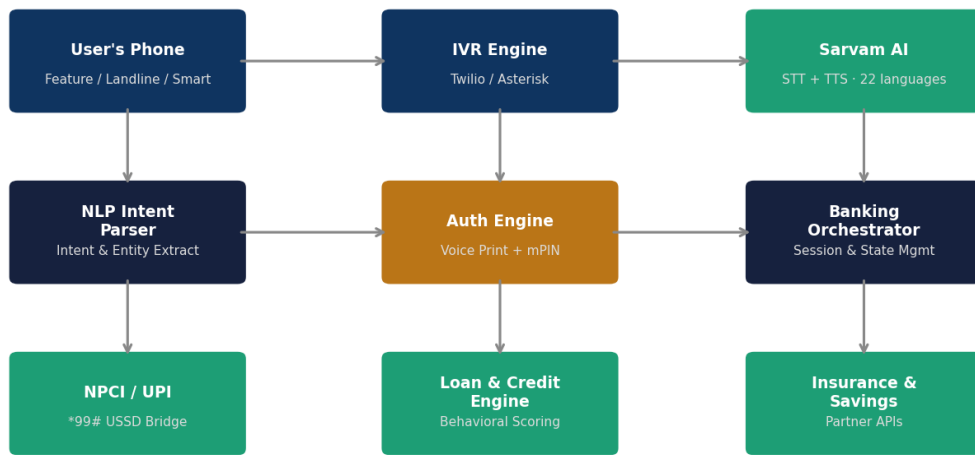


Figure 2: VaaniPay system architecture: five-layer design from telephony to financial services

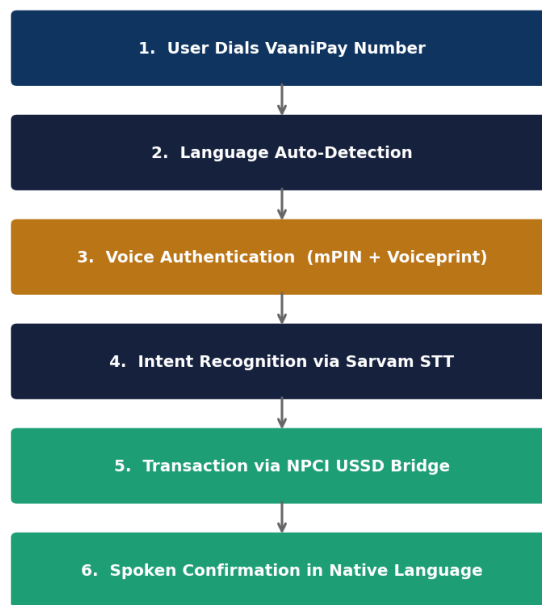


Figure 3: IVR call flow for a UPI payment transaction

Key Features

VaaniPay handles all UPI transactions via voice - user speaks the beneficiary's name or number and the amount and transaction is executed without any need of any button press or keypad. User can inquire about balance, mini statements and transfer money via voice commands only. The platform supports fully voice-driven applications for micro loans between 500 to 25,000 (with instant decisions taken based on the user's in-app behavioral credit score). It also supports fully voice-activated daily health insurance, accident insurance,

crop insurance, recurring deposit, fixed deposit and allows users to deposit funds via voice. The user can also get their EPF/NPS balance from EPFO/PFRDA via respective API integration.

Most unique part is the in-app behavioral credit scoring mechanism to serve the 'New to Credit' (NTC) customers who have never had a formal credit history. Score is derived based on five categories of signals - consistency of transactions, timely repayment of previous micro loans, savings consistency, level of platform usage and strength of referral network. Score is calculated on the standard 300-900 credit scoring scale (used by Indian credit bureaus). And interestingly, when a user inquires about their credit score, VaaniPay reads out the most significant factors contributing to the score back to the user, making the user aware of how to improve their credit score as well as grant credit access to them. Loans are sanctioned and interest rate is regulated and depends on credit score (for score >650 they can avail higher loan amount and lower interest rates).

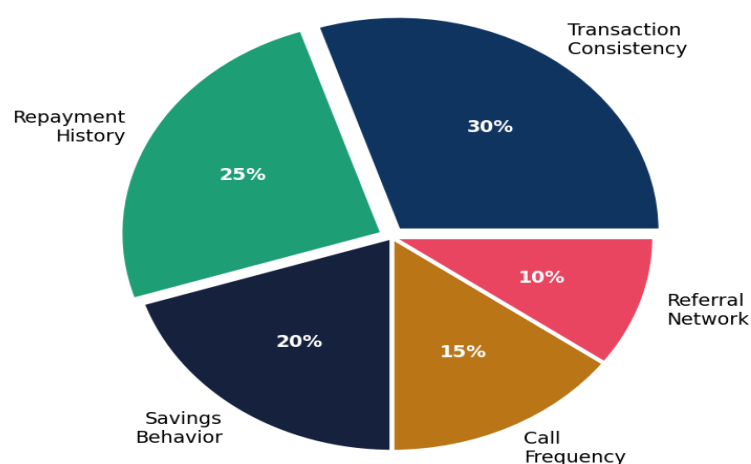


Figure 4: Behavioral credit scoring — weightage of five signal categories

Table I: VaaniPay Service Portfolio

Service	Description	Infrastructure	Target User
UPI Payment	Send/receive money by speaking name or number	NPCI *99# USSD Bridge	All users
Micro-Loan	₹500–₹25,000 instant loans via voice	NBFC partner APIs	Gig workers, farmers
Micro-Insurance	Health, accident & crop coverage by voice	Digit / Acko APIs	Rural households
Credit Score	Behavioral score from in-platform history	Proprietary scoring engine	New-to-credit users
FD / Savings	Open fixed/recurring deposits by voice	Small Finance Bank APIs	Low-income savers
PF / NPS	Check EPF and NPS balance via voice	EPFO / PFRDA APIs	Formal sector workers

Tech Stack

Speech intelligence layer: The STT and TTS for all 22 Indian languages were completely built in-house at Sarvam AI specifically for Indic speech and not an adapted version of models that have been fine tuned on English data. As a result they perform significantly better in telephony settings. **IVR and telephony layer:** The solution uses either Twilio or Asterisk (depending on the deployment requirements of cloud vs. On-prem) for PSTN and GSM call routing where the internet is not required for the caller's phone.

Backend layer: Built on Node.js and Python micro services. **NLP intent parser:** Custom model fine tuned on Indic languages with a finance lexicon. **Data storage:** User profiles, transaction data, behavioral credit data are stored in PostgreSQL. User session is stored in Redis to maintain state through multi-turn voice interactions within a call. **Payment execution:** Payments are executed through the NPCI *99# USSD bridge. Loan disbursement is achieved via NBFC partner APIs. Insurance offerings can be integrated with InsurTech partners licensed by IRDAI like Digit, Acko.

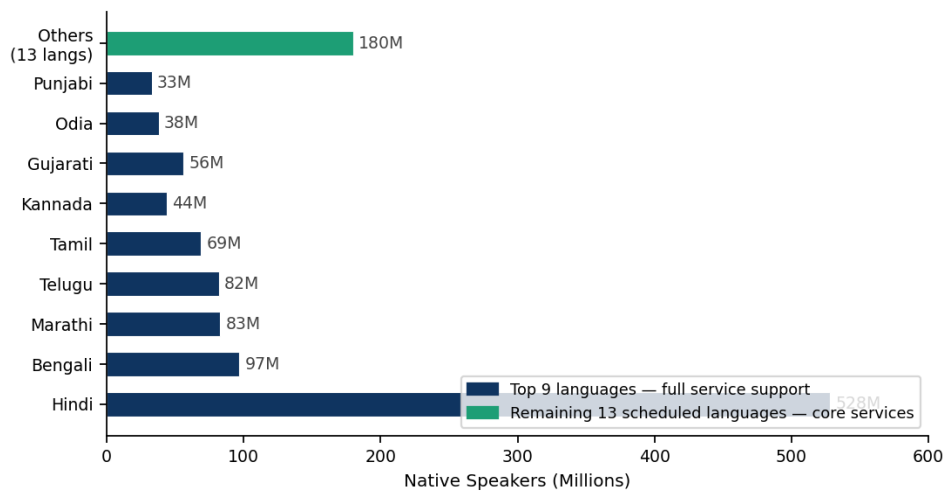


Figure 5: Native speaker population for each of the 22 scheduled languages supported by VaaniPay

Table II: Technology Stack

Layer	Technology	Purpose
Speech AI	Sarvam AI STT + TTS	Speech recognition and synthesis across 22 Indic languages
Telephony	Twilio / Asterisk	IVR engine, PSTN/GSM call routing, DTMF fallback
Intent Parsing	Custom NLP model	Financial intent classification and entity extraction
Authentication	Voice biometrics + mPIN	Multi-factor identity verification per call
Backend	Node.js + Python microservices	Banking orchestration, session and state management

Database	PostgreSQL + Redis	Persistent user data and in-call session state
Payments	NPCI *99# USSD Bridge	UPI transaction execution over feature phone networks
Finance APIs	NBFC + InsurTech + EPFO	Loan disbursement, insurance products, PF/NPS enquiry

Feasibility

The technology stack has been validated, with Sarvam AI's models performing at production-level accuracy under telephony circumstances and NPCI's *99# system processing millions of USSD transactions per day. The regulatory framework is also in place-USSD based UPI is already being implemented under RBI regulations from 2014, voice-based KYC is in place from 2020 (V-CIP by RBI), micro-lending to NBFC partners is under current frameworks, and micro-insurance with IRDAI licensed partners is already a model (e.g. PMJBY/PMSBY).

Commercially, VaaniPay is viable, without any need for grants and subsidies. Revenue is generated via transaction fees of 0.5%-1% on the payments, and loan origination fees on loans distributed via NBFC partners. B2B licensing of the voice banking API to NBFCs, banks and NGOs to act as a white-labeled voice banking platform will provide enterprise revenue to the business in addition to consumer revenue streams. The existing partnership with the government via Jan Dhan Yojana and PM Fasal Bima provides a ready distribution pipeline into the exact segment for whom VaaniPay was built.

Table III: VaaniPay vs. Existing Solutions

Feature	GPay / PhonePe	NPCI *99#	M-Pesa (Kenya)	VaaniPay
Smartphone required	Yes	No	No	No
Internet required	Yes	No	No	No
Voice / NLP interface	No	No	No	Yes
Languages supported	8–10	9	1	22
Micro-loans	No	No	Limited	Yes
Micro-insurance	Limited	No	No	Yes
Credit score building	No	No	Limited	Yes
Literacy requirement	High	Medium	Low	None

Impact

VaaniPay's immediate outcome is access – 250 million feature phone owners given first formal digital finance journey. The second and more significant outcome is structural. Each transaction creates a behavioral credit profile, that in 90 days, for a day-wage earner with no existing credit score, qualifies them for a micro-loan at a regulation rate-saving them from 24-60% annually from lenders-and for each activated micro-insurance, a farmer or domestic worker has protection against the health and accident shocks that could erase 6 months of saving.

Women in rural communities are the primary users, excluded by default in smartphone dominated digital finance. A zero-literacy, zero-smartphone, native-language interface drops every wall that's excluded them traditionally from finance. In scale, VaaniPay doesn't merely serve unbanked India; it formally includes them-Users with a credit history from VaaniPay are ready for mainstream banks products, the platform acts as a floor, not a ceiling.

There are 1.15 Billion phone subscribers in India. Every one of those is in VaaniPay's addressable market. The architecture scales seamlessly to Bangladesh, Nepal, and Sri Lanka upon expansion of Sarvam Language Models, and is replicable in any emerging market that exhibits feature phone dominance. The problem VaaniPay is addressing is not India's unique problem, it is the developing world's dominant financial inclusion problem.